

CHM 120

Hess's Law

Final Report

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Lesson	Hess's Law
Institution	Rock Valley College
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Course	CHM 120
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Test Your Knowledge

Match each term with the best description.

States the total change in enthalpy is independent of the pathway	1 Hess's Law
The unit of measure for change in the enthalpy according to the International System of Units	2 Joule
Reaction that absorbs heat from the surrounding environment	3 Endothermic
The study of heat that accompanies chemical reactions	4 Thermochemistry
Reaction that releases heat to the surrounding environment	5 Exothermic
The internal energy of a system that includes heat plus the product of pressure and volume	6 Enthalpy

Categorize each statement as representing an exothermic or endothermic reaction.

Exothermic		Endothermic	
1	Temperature increases	2	Temperature decreases
The change in enthalpy is a negative number		Heat is absorbed from surroundings	
Heat is released to surroundings		The change in enthalpy is a positive number	

Match term with the correct equation.

= Moles of solute (mol)/ volume of solution (L)	1	Molarity
= Final temperature - Initial temperature	2	Change in temperature
= (mass of solution) x (specific heat) x ΔT	3	Heat gain or loss
$\frac{\text{Experimental value} - \text{Theoretical value}}{(\text{Theoretical value})} \times 100\%$	4	Percent Error
= $H_{\text{products}} - H_{\text{reactants}}$	5	Change in enthalpy

Exploration

When an ____ reaction occurs in an aqueous solution, heat will be released to the water.

- ☐ endothermic
☒ exothermic

Enthalpy is the external energy of the system, E , which includes heat plus the product of pressure and volume.

- ☐ True
- ☒ False

The change in enthalpy of a reaction is equal to the enthalpy of the reactants minus the enthalpy of the products.

- ☐ True
- ☒ False

When a reaction is exothermic, the enthalpy of the products is ____ the reactants.

- ☐ greater than
- ☒ less than
- ☐ equal to

Hess's law states that the total change in enthalpy is independent of the pathway.

- ☒ True
- ☐ False

Exercise 1

What were the maximum temperatures for each of the reactions? Use the scatter plots in Graph 1, Graph 2, and Graph 3 to support your answers.

B / <u>U</u>	15 Word(s)
Reaction 1: 29.4 Degrees C Reaction 2: 20.8 Degrees C Reaction 3: 23.9 Degrees C	

Calculate the heat loss or heat gain of the 3 solution mixtures (q_{rxn}). Then calculate the ΔH for each reaction. Write the calculations on a sheet of paper showing all of your work. Take a photo of your work for each reaction and upload the images into Photo 1, Photo 2, and Photo 3.

B / U

77 Word(s)

I am a little stumped on how to calculate the change in H of each reaction. I got confused reading the notes regarding how to add components together. Do I have to find the molar mass of each reactant before then adding everything together? Again, it looks straight forward, but I am lost. I just didn't want to upload a bunch of nonsense because looking at what I do have written down, things look ridiculous and messy.

Use Hess's law and your determination of ΔH for the first 2 reactions to determine ΔH for the third reaction:



B / U

30 Word(s)

Change in H for the third reaction is equal to the change in H of the first reaction plus the opposite of the change in H in the second reaction:

Compare the results of question 3 with the experimental results of the reaction:



B / U

19 Word(s)

I was having a problem with question two, which effects question three, and ultimately effecting question four (this question).

Use the thermodynamic quantities given below to calculate the theoretical ΔH for this reaction:



- ΔH°_f for NH_3 (aq) = - 80.29 kJ/mol
- ΔH°_f for HCl (aq) = - 167.2 kJ/mol
- ΔH°_f for NH_4^+ (aq) = - 132.5 kJ/mol
- ΔH°_f for Cl^- (aq) = - 167.2 kJ/mol

B /

4 Word(s)

Change in H = -0.55J/mol

What is the percent error of experimental results when using the theoretical value calculated in Question 5?

B /

37 Word(s)

I'm really not sure if what I did in question five is correct. I'm also unsure of what the actual yield is that needs to be considered for the equation in order to calculate the percent error.

Were the reactions performed in this laboratory exercise exothermic or endothermic? How could you determine this?

B /

21 Word(s)

Exothermic because energy was expelled and decreased as time went on. The graphs all show decrease in energy as time continues

Define Hess's law. Did your experimental results support Hess's law? Use your data to explain your answer.

B /

22 Word(s)

Hess's Law states that the total enthalpy change for a reaction is the sum of all changes (steps) that a reaction goes through.

What are some possible sources of error in this experiment?

B / <u>U</u>	49 Word(s)
Changing the environment of the components while they are trying to establish equilibrium could take longer. Maybe if tap water was used instead of distilled water, things could have been effected. Possibly the amount of the components going into the reaction could give different results, but I doubt it.	

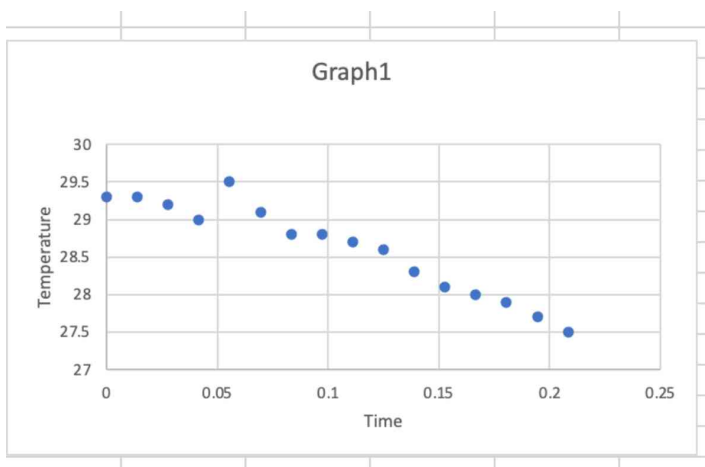
Data Table 1: Temperature (°C) of the Solution (NaOH + HCl)

Initial Temperature of NaOH (°C)	20.5
Initial Temperature of HCl (°C)	20.8
Average Initial Temperature of Two Separate Solutions (°C)	20.7
Highest Temperature of Mixture (May Be Extrapolated from Graph) (°C)	29.3
Change in Temperature of Mixture (°C), ΔT	8.6

Data Table 2: Temperature of Solution Every 20 Seconds

Time after mixing (min:sec)	Temperature (°C)
Initial	29.3
0:20	29.2
0:40	29.0
1:00	29.5
1:20	29.1
1:40	28.8
2:00	28.8
2:20	28.7
2:40	28.6
3:00	28.3
3:20	28.1
3:40	28.0
4:00	27.9
4:20	27.7
4:40	27.5
5:00	27.4

Graph 1: Reaction 1 (NaOH + HCl)



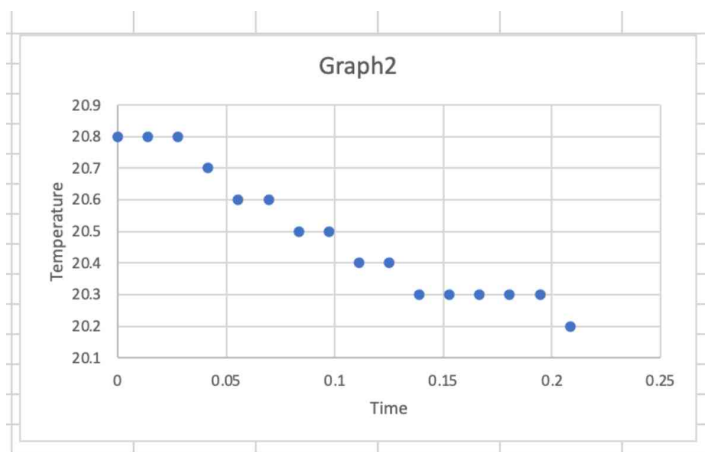
Data Table 3: Temperature (°C) of the Solution (NaOH + NH₄Cl)

Initial Temperature of NaOH (°C)	20.1
Initial Temperature NH ₄ Cl (°C)	20.7
Average Initial Temperature (°C)	20.4
Highest Temperature of Mixture (May Be Extrapolated from Graph) (°C)	20.8
Change in Temperature of Mixture (°C), ΔT	.40

Data Table 4: Temperature of Solution Every 20 Seconds

Time after mixing (min:sec)	Temperature (°C)
Initial	20.8
0:20	20.8
0:40	20.7
1:00	20.6
1:20	20.6
1:40	20.5
2:00	20.5
2:20	20.4
2:40	20.4
3:00	20.3
3:20	20.3
3:40	20.3
4:00	20.3
4:20	20.3
4:40	20.3
5:00	20.2

Graph 2: Reaction 2 (NaOH + NH₄Cl)



Data Table 5: Temperature (°C) of the Solution (NH₃ + HCl)

Initial Temperature of NH ₃ (°C)	20.9
Initial Temperature HCl (°C)	20.6
Average Initial Temperature (°C)	20.8
Highest Temperature of Mixture (May Be Extrapolated from Graph)(°C)	23.9
Change in Temperature of Mixture (°C), ΔT	3.1

Data Table 6: Temperature of Solution Every 20 Seconds

Time after mixing (min:sec)	Temperature (°C)
Initial	23.9
0:20	23.8
0:40	23.8
1:00	23.8
1:20	23.8
1:40	23.7
2:00	23.6
2:20	23.4
2:40	23.3
3:00	23.2
3:20	23.1
3:40	23.1
4:00	23.0
4:20	23.0
4:40	22.9
5:00	22.9

Graph 3: Reaction 3 (NH₃ + HCl)

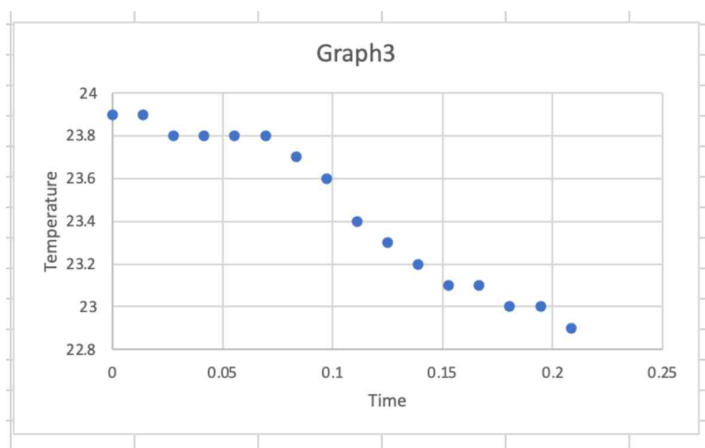


Photo 1: Reaction 1: NaOH + HCl Calculations

Upload an image

Photo 2: Reaction 2: NaOH + NH₄Cl Calculations

Upload an image

Photo 3: Reaction 3: NH₃ + HCl Calculations

Upload an image

Competency Review

In an endothermic reaction ____.

- ☐ heat is released and temperature increases
- ☐ heat is released and temperature decreases
- ☐ heat is absorbed and temperature decreases
- ☒ heat is absorbed and temperature increases

A chemical reaction that releases heat to the surrounding environment is considered ____.

- ☐ endothermic
- ☒ exothermic

The amount of heat given off or absorbed by a chemical reaction is called ____.

- ☐ an exothermic reaction
- ☒ the change in enthalpy
- ☐ a Joule
- ☐ the product of pressure and volume

Enthalpy is denoted by the ____ symbol.

- ☒ H
- ☐ J
- ☐ ΔE
- ☐ F

The change in enthalpy of a reaction is equal to the enthalpy of the ____ minus the enthalpy of the ____.

- ☒ products; reactants
- ☐ reactants; products
- ☐ None of the above

When an ____ reaction occurs the products have a greater enthalpy than the reactants, whereas during an ____ reaction the enthalpy of the products is less than the enthalpy of the reactants.

- ☐ exothermic; endothermic
- ☒ endothermic; exothermic

Hess's Law indicates that if a reaction takes place in one or multiple steps, the change in enthalpy for the overall process must be the sum of the change in enthalpy of the constituent reactions.

- ☒ True
- ☐ False

Calculate the percent error of a chemical reaction with a theoretical value of -45.68 kJ/mol and an experimental value of -54.78 kJ/mol.

- ☐ 20.10%
- ☒ 16.61%
- ☐ 19.92%
- ☐ 21.37%

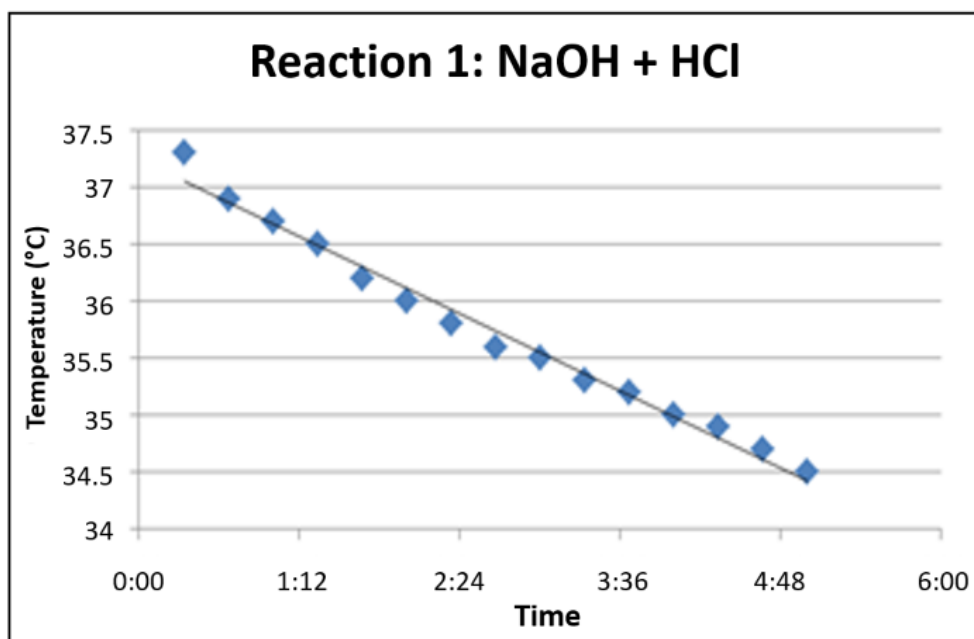
What is the ΔH of a reaction where the mass of the solution is 18.7 g at a specific heat of 5.008 J/g°C and the system experiences a change in temperature of 7.2°C?

- ☒ 674.27 J
- ☐ -674.27 J
- ☐ -67.42 J
- ☐ 67.42 J

If an experiment has a q_{rxn} of 518.07 J, calculate the change in enthalpy of the reaction if there were 0.04 mol of product in the solution.

- ☒ 12.95 kJ/mol
- ☐ -12951.75 kJ/mol
- ☐ -12.95 kJ/mol
- ☐ 7.72 kJ/mol

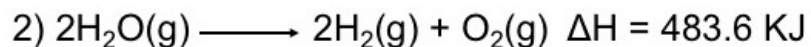
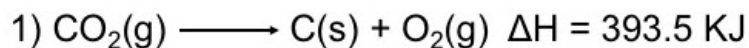
In an experiment, NaOH and HCl are mixed. Based on the graph, what is the change in temperature of the mixture if the initial temperature of the NaOH was 23.0°C and the initial temperature of the HCl was 25.2°C?



- ☐ 1.3°C
- ☐ 12.7°C
- ☒ 13.2°C
- ☐ 10.4°C

Extension Questions

Worldwide, scientists are working on different options for clean energy to replace the burning of fossil fuels. One option that could potentially be successful is the use of hydrogen gas. This gas can be generated through a reaction between carbon and water.



Based on what you have learned in this lab, use the chemical equation and Hess's law to determine the heat of formation and if this reaction is exothermic or endothermic.

The reaction is endothermic. Change in H is roughly 438.6 KJ